# CCDAK Knowledge Domains

## Application Design

- Kafka's command line tools
- Pub/Sub and Streaming
- Overall Kafka's architecture and design
- Message metadata
- System metrics
- Message Key selection (choices and factors)
- Message schema management

## Development

- Clients: Producer and Consumer, concepts and functions
- Client troubleshooting/debugging
- Performance, throughput, latency, scaling
- Message order and delivery guarantees
- Serialization/Deserialization
- Consumer offset management
- Consumer groups, partition assignment, partition rebalances
- Data retention strategies and implications
- Topic co-partitioning

## Deployment/Testing

- Securing your data
- Monitoring and troubleshooting clients
- Kafka stream features and uses cases
- Essential component parts of Kafka Streams application
- KSQL/ksqlDB features and use cases:
- Essential elements of KSQL/ksqlDB env